



Paterna, September 13rd, 2026

Wellcome Leap Resistance Networks Program

Re: Support for “Universal Properties of Plasmid Evolution and Robust Risk Forecasting for Preemptive pARG Intervention”

Dear Members of the Review Committee:

I am writing in support of the proposal submitted by Dr. Philip J. Gerrish to the Wellcome Leap Resistance Networks program.

I have known Phil since we worked together in Prof. Richard Lenski's laboratory early in our careers. He has remained a close colleague and personal friend. Our scientific collaboration has included publications in *Science*, the *Journal of Virology*, and *PLoS Computational Biology* on clonal interference, estimation of viral mutation rates, and the effects of recurrent population bottlenecks on genome evolution. I therefore know his work directly and can assess his scientific judgment and ability to connect evolutionary theory with empirical data.

The proposed treatment of resistance plasmids as evolving transmissible elements is well founded. Plasmids and viruses both experience repeated changes of biological context, and their persistence depends on interacting processes operating within and between hosts. Our previous work has shown that mutation, selection, bottlenecks, and transmission can produce evolutionary outcomes that are not apparent when these processes are considered separately. This experience is directly relevant to the proposed study of plasmid transmission and persistence.

The proposal continues a longstanding feature of Phil's research: deriving general results from evolutionary theory and expressing them as quantitative, falsifiable predictions. His early theoretical work on competing beneficial mutations led to an experimentally testable account of clonal interference that we subsequently examined in RNA virus populations. The present project applies the same combination of theory, statistical analysis, and empirical validation to an important problem in antimicrobial resistance.

Phil has the theoretical background, independence, and persistence required to carry out this work. I support the proposal without reservation.

Sincerely,

Firmado por ELENA FITO SANTIAGO FRANCISCO -
DNI ***9610** el día 13/09/2026 con un
certificado emitido por AC Sector Público